Aravind Sudheer

India | aravind.sudheer01@gmail.com
linkedin.com/in/aravindsudheer/ | github.com/aravindsudheer01-arch

PROFILE

Robotics and mechatronics engineering graduate with professional ABB automation experience and project-based Python, AI application and computer vision skills. Developed applications involving APIs, structured outputs, metadata processing and interactive interfaces.

TECHNICAL SKILLS

Programming and data: Python, SQL, pandas, NumPy, SQLite

AI applications: LangChain, Pydantic, OpenAI API, Gmail API, Streamlit

Automation: ABB robots, RAPID, RobotStudio, PLC signal verification, WinCC SCADA

SELECTED PROJECTS

AI Research Assistant — Tool-Calling Prototype
Python, LangChain, OpenAI API, Pydantic

- Developed a CLI assistant integrating web search, structured model responses and timestamped report storage.
- Filtered reported source URLs against observed search results; added helper tests for evidence reconciliation and file persistence.

Content-Based Movie Recommendation System
Python, pandas, scikit-learn, Streamlit

- Built a Streamlit recommender with a combined-text TF-IDF baseline and weighted similarity across five metadata features.
- Added schema checks, duplicate-title handling and feature-level explanations; documented limitations and synthetic tests.

Driver Monitoring System — Computer Vision Prototype
Python, OpenCV, MediaPipe, NumPy

- Integrated pretrained facial landmarks with eye/mouth geometry and timed logic for sustained observations and audio alerts.
- Separated webcam, monitoring and alert modules; included synthetic tests without claiming real-driving reliability.

PROFESSIONAL EXPERIENCE

Junior Robotics Engineer
Simpliforge Creations | Hyderabad, India | Dec 2025 – May 2026

- Programmed and simulated ABB six-axis robots using RAPID and RobotStudio for concrete 3D printing.
- Configured and calibrated workobjects, tool data and end-effectors before robot-program execution.
- Troubleshot robot-motion and program-execution issues during live printing; monitored PLC signals and process parameters.

Design & Application Engineer (Intern)
Automation Systems and Services | Kuala Lumpur, Malaysia | Nov 2023 – Feb 2024

- Supported industrial automation with WinCC SCADA layouts, panel wiring, PLC signal checks and commissioning assistance.
- Assisted system validation and technical documentation; reorganized packing and setup workflows for time-sensitive deployments.

EDUCATION

Bachelor of Engineering (Honours) — Robotics and Mechatronics Engineering

Monash University | Graduated July 2025